

Let, WLOG, previous P1 be 1. $\therefore$ P2 is 1, $\therefore$ Previous P2 is 0, according to given condition. $\therefore$ P3 is 1, $\therefore$ Previous P3 is 1, according to given condition. Repeating the argument, we get: previous P4 is 0, previous P5 is 1, previous P6 is 0, previous P7 is 1, previous P8 is 0, previous P9 is 1.

Now, $\therefore$ P1 is 1, $\therefore$ previous P1 is 0. This is a contradiction to our assumption that previous P1 is 1.

We can arrive at a similar contradiction if we had assumed previous P1 as 0.

$\therefore$ We can never get 9 0's at the end. $\square$

29. Formulate a conjecture about the decimal digits that appear as the final decimal digit of the fourth power of an integer. Prove your conjecture using a proof by cases.

Solution: We shall first check the final decimal digits of the fourth powers of each decimal digit.

$0^4 = 0 \rightarrow$ Final digit 0 $1^4 = 1 \rightarrow$ Final digit 1 $2^4 = 16 \rightarrow$ Final digit 6
 $3^4 = 81 \rightarrow$ Final digit 1 $4^4 = 256 \rightarrow$ Final digit 6 $5^4 = 625 \rightarrow$ Final digit 5 $6^4 = 1296 \rightarrow$ Final digit 6
 $7^4 = 2401 \rightarrow$ Final digit 1 $8^4 = 4096 \rightarrow$ Final digit 6 $9^4 = 6561 \rightarrow$ Final digit 1

Claim: For any arbitrary integer n , the final decimal digit of n^4 is one of 0, 1, 6, 5.

Proof: Any arbitrary integer n can be written as $10x+y$, where x is the portion of n containing every digit apart from the last, and y is the last digit. The following cases arise:

Case 1: $y=0 \therefore n=10x+0=10x$ $n^4=(10x)^4=10000 \cdot x^4$

We know the fact that any number on multiplication with a multiple of 10 ends with 0. $\therefore n^4$ ends with 0. $\square$